Supplementary Material

A computational model of the interactions between formate and insulin metabolism

Alexei Vazquez

Detailed model equations, parameters, calibration and numerical methods

This document gives the full technical specification of every model used in the main text: state variables, rate laws, ODE balances, parameter values with provenance, calibration procedures, and numerical methods. It is intended as a complete and reproducible reference; the main text is kept to the biological narrative and points here for detail. The model is implemented as an object-oriented class hierarchy in `model.py` — a `MetabolicCell` base (shared numerics and rate-law primitives) subclassed by `BetaCellModel`, `AdipocyteModel` and `EyeModel`, with `WholeBody`, `FormateDiabetes`, `CoupledModel` and its gut-microbiome extension `GutUrateCycle` containers, each carrying an explicit parameter dataclass. All code, calibrated outputs (`data/*.csv`) and figure scripts are in the accompanying repository.

Contents

S1 Conventions and numerical methods

Units. Time is in hours; metabolite concentrations in mM; insulin and lipid pools in arbitrary anabolic units (a.u.); plasma insulin in a.u. Blood formate is reported in mM and, for clinical comparison, in mg/L using the formate molecular weight 46.03 (mg/L = mM $\times 46.03$); blood urate in mg/dL using 16.8.

Provenance tags. Every parameter carries one of four tags. **[PUB]** inherited from the previously published Formate–NUDT5 purine/energy core (Vazquez, submitted); **[FIT]** fitted here by least squares to the calibration targets of §S2.8; **[LIT]** taken from a specific literature measurement; **[ASM]** assumed / order-of-magnitude estimate not individually measured. Tags are given in the parameter tables.

Integration. All ODEs are integrated with an implicit backward-differentiation (BDF) solver (`scipy.integrate.solve_ivp`) with dense output. Tolerances are stated per model.

Steady states. Unless noted, a steady state is obtained by pre-integrating to a physiologically reasonable basin and then polishing the residual in log-space with `least_squares` (variables $z = \log y$, residual $\dot{y}(e^z)/(e^z + 10^{-6})$, so states stay positive), to residuals $< 10^{-13}$ mM/h. The one exception is the regulated eye feed-forward (§S6.2), where a steady state is obtained by integration to the stable attractor rather than root-polishing, for reasons made precise in §S6.3.

S2 The pancreatic β-cell model

The core is a fifteen-state kinetic model of the coupled energy, one-carbon, purine, mTORC1 and insulin-synthesis network of a glucose-responsive β-cell. The purine/energy backbone and the PPAT–NUDT5 "glue" calibration are inherited from the Formate–NUDT5 model; the insulin-synthesis, secretion and glucose-sensing layers are added and fitted here.

S2.1 State variables

All states are floored at 10^{-12} before rate evaluation. A common dilution constant $\mu = 1/240$ h^{-1} ($\approx$ ten-day pool renewal) applies to most pools; `form`, `cho`, `hx` and `urate` carry no dilution term.

S2.2 Energy metabolism

Glucose G sets a glucokinase gate and a fuel level; ATP is produced by glycolysis and oxidative phosphorylation (each Michaelis–Menten in ADP), consumed by an ATPase and by anabolism,

Table S1: The fifteen β-cell state variables (array order), with initial condition Y_0.

i	symbol	meaning	units	$Y_{0,i}$
0	`atp`	ATP	mM	3.0
1	`adp`	ADP	mM	0.3
2	`amp`	AMP	mM	0.03
3	`form`	intracellular formate	mM	0.3
4	`cho`	10-formyl-THF (10-CHO-THF)	mM	0.01
5	`prpp`	PRPP	mM	0.1
6	`gar`	GAR (glycinamide ribonucleotide)	mM	0.02
7	`aic`	AICAR	mM	0.02
8	`imp`	IMP (purine hub)	mM	0.05
9	`samp`	adenylosuccinate (S-AMP)	mM	0.02
10	`gnp`	guanine nucleotides	mM	1.0
11	`hx`	hypoxanthine/xanthine	mM	0.02
12	`urate`	urate	mM	0.05
13	`proins`	pro-insulin	a.u.	1.0
14	`ins`	mature insulin (granule pool)	a.u.	5.0

and the adenine nucleotides equilibrate through adenylate kinase:

$$gk = \frac{G^{h_{gk}}}{K_{gk}^{h_{gk}} + G^{h_{gk}}}, \qquad \text{fuel} = \text{BASALFUEL} + (1 - \text{BASALFUEL})\, gk, \tag{S1}$$

$$\text{cox} = \Big[1 + (\text{form}/K_i^{\text{cox}})^2\Big]^{-1} \text{ (else 1)}, \qquad v_{glyc} = e_g \,\text{fuel}\, \frac{adp}{E_g + adp}, \tag{S2}$$

$$v_{ox} = e_o \,\text{fuel}\, \frac{adp}{E_o + adp}\,\text{cox}, \qquad v_{atpase} = a\, \frac{atp}{A + atp}, \tag{S3}$$

$$r_{ak} = k_{ak}\Big(adp^2 - \frac{amp \cdot atp}{K_{ak}}\Big), \qquad v_{atp}^{\text{prod}} = v_{glyc} + v_{ox}. \tag{S4}$$

The complex-IV factor cox is a formate inhibition of oxidative phosphorylation used only in the (optional, weakly evidenced) methanol complex-IV scenario; it is *disabled by default* (K_i^{cox} = None $\Rightarrow$ cox = 1) and is switched off entirely in the main-text eye analyses.

S2.3 One-carbon entry and de novo purine synthesis

Formate is charged onto tetrahydrofolate and drives the two formyl-transfer steps of purine ring assembly; the committed step (PPAT) is gated by the AMP/PRPP "glue" term θ:

$$\text{thf} = F_{tot} - cho, \qquad v_{fthfs} = h\, \frac{form}{H+form}\, \frac{\text{thf}}{F_{tot}}, \qquad \text{onec} = \frac{cho}{K_c+cho}, \tag{S5}$$

$$\theta = f_N\, \frac{1+(amp/K_a)^n}{1+(amp/K_a)^n+prpp/K_{dP}}, \qquad v_{ppat} = P_{max}\, \frac{prpp}{K_{prpp}+prpp}\,(1-\theta), \tag{S6}$$

$$v_{gart} = g_{max}\, \frac{gar}{K_g+gar}\,\text{onec}, \qquad v_{atic} = t_{max}\, \frac{aic}{K_t+aic}\,\text{onec}. \tag{S7}$$

$f_N \in [0,1]$ scales the NUDT5 glue strength ($f_N = 0$ disables the AMP/PRPP feedback). The glue parameters K_a, n are fitted (§S2.8) to the Witus *et al.* PRPP IC_{50} vs. AMP data.

S2.4 Purine interconversion, degradation and urate

With adenylate ratio $adr = atp/adp$:

$$v_{imp\to amp} = k^{\text{syn}}_{amp}\, imp\, adr, \quad v_{samp\to amp} = V^{samp}_{max} \frac{samp}{K_{samp} + samp}, \quad v_{imp\to gmp} = k^{\text{syn}}_{gmp}\, imp\, adr, \tag{S8}$$

$$v^{\text{deg}}_{imp} = k^{\text{deg}}_{imp}\, imp, \qquad v^{\text{deg}}_{amp} = k^{\text{deg}}_{amp}\, amp, \qquad v^{\text{turn}}_{ade} = k^{\text{turn}}_{ade}\, atp, \tag{S9}$$

$$v^{\text{deg}}_{gnp} = k^{\text{deg}}_{gnp}\, gnp, \qquad v_{xo} = k_{xo} \frac{hx}{K_{hx} + hx}, \qquad v^{\text{sec}}_{urate} = k_{urate}\, urate, \tag{S10}$$

and a guanine-nucleotide usage $v^{\text{use}}_{gnp} = k^{\text{use}}_{gnp}\, gnp$.

S2.5 mTORC1 and insulin synthesis

mTORC1 senses a guanine-weighted purine pool and the adenylate energy charge; insulin transcription is glucose-driven; peptide synthesis multiplies the two; maturation and an ATP/ADP-triggered secretion complete the granule cycle:

$$P_{\text{pool}} = W_{\text{ATP}}\, atp + gnp, \qquad ec = \frac{atp + 0.5\, adp}{atp + adp + amp}, \tag{S11}$$

$$\text{mtor} = \text{mtor\_mult} \cdot \frac{P^{h_{mp}}_{\text{pool}}}{K^{h_{mp}}_{mp} + P^{h_{mp}}_{\text{pool}}} \cdot \frac{ec^{h_{ec}}}{K^{h_{ec}}_{ec} + ec^{h_{ec}}}, \qquad g_{tx} = B_{TX} + (1 - B_{TX}) \frac{G^{h_G}}{K^{h_G}_{G} + G^{h_G}}, \tag{S12}$$

$$v^{\text{syn}}_{\text{ins}} = k_{ins}\big(B_M + (1 - B_M)\, \text{mtor}\big)\, g_{tx} \frac{aa}{K_{aa} + aa}, \qquad v_{mat} = k_{mat}\, proins, \tag{S13}$$

$$\text{trig} = B_{TR} + (1 - B_{TR}) \frac{(atp/adp)^{h_{tr}}}{K^{h_{tr}}_{tr} + (atp/adp)^{h_{tr}}}, \qquad v_{sec} = k_{sec}\, ins\, \text{trig}. \tag{S14}$$

The ATP cost of anabolism is $v^{\text{cost}}_{atp} = \text{cost}_{ins}\, v^{\text{syn}}_{\text{ins}} + \text{cost}_{pur}\, v_{ppat}$. The multiplier mtor_mult (default 1; 0 = Raptor knockout) and the mTOR-independent floor B_M set the mTORC1 dependence of insulin content (fitted to Ni *et al.*, §S2.8).

S2.6 ODE balances

With the fluxes above (and the override arguments f_{mito}, the mitochondrial formate production; f_N; mtor_mult; K_i^{cox} described in-line), the state derivatives are:

$$\dot{atp} = v_{atp}^{\text{prod}} - v_{atpase} - v_{atp}^{\text{cost}} + r_{ak} - v_{ade}^{\text{turn}} - \mu\, atp, \tag{S15}$$

$$\dot{adp} = -v_{atp}^{\text{prod}} + v_{atpase} + v_{atp}^{\text{cost}} - 2r_{ak} - \mu\, adp, \tag{S16}$$

$$\dot{amp} = r_{ak} + v_{samp\to amp} - v_{amp}^{\text{deg}} - \mu\, amp, \tag{S17}$$

$$\dot{form} = f_{\text{mito}} + k_F\,(\text{form}_x - form) - v_{fthfs}, \tag{S18}$$

$$\dot{cho} = v_{fthfs} - v_{gart} - v_{atic}, \tag{S19}$$

$$\dot{prpp} = S_{prpp} - v_{ppat} - k_o^{prpp}\, prpp - \mu\, prpp, \tag{S20}$$

$$\dot{gar} = v_{ppat} - v_{gart} - (\mu + k_{\text{deg}}^{\text{int}})\, gar, \tag{S21}$$

$$\dot{aic} = v_{gart} - v_{atic} - (\mu + k_{\text{deg}}^{\text{int}})\, aic, \tag{S22}$$

$$\dot{imp} = v_{atic} - v_{imp\to amp} - v_{imp\to gmp} - v_{imp}^{\text{deg}} - \mu\, imp, \tag{S23}$$

$$\dot{samp} = v_{imp\to amp} - v_{samp\to amp} - \mu\, samp, \tag{S24}$$

$$\dot{gnp} = v_{imp\to gmp} - v_{gnp}^{\text{deg}} - v_{gnp}^{\text{use}} - \mu\, gnp, \tag{S25}$$

$$\dot{hx} = v_{imp}^{\text{deg}} + v_{amp}^{\text{deg}} + v_{ade}^{\text{turn}} + v_{gnp}^{\text{deg}} - v_{xo}, \tag{S26}$$

$$\dot{urate} = v_{xo} - v_{urate}^{\text{sec}}, \tag{S27}$$

$$\dot{proins} = v_{\text{ins}}^{\text{syn}} - v_{mat} - \mu\, proins, \tag{S28}$$

$$\dot{ins} = v_{mat} - v_{sec} - \mu\, ins. \tag{S29}$$

Here form_x is the blood formate seen by the cell and f_{mito} (default FMITO) the endogenous mitochondrial formate; lowering f_{mito} represents MTHFD1L loss / mitochondrial formate deficiency.

S2.7 Parameters

Table S2: β-cell parameters. Derived values expanded numerically (m=84, e=2m=168, v_{pur}=0.30).

symbol	value	meaning	tag
		Energy / adenine nucleotides	
K_{gk}	8.0 mM	glucokinase half-saturation	[LIT]
h_{gk}	1.7	glucokinase Hill coefficient	[LIT]
m	84 mM/h	maintenance ATP turnover	[PUB]
e	168 mM/h	total ATP-cycling scale ($2m$)	[PUB]
e_g	100.8 mM/h	max glycolytic ATP rate ($0.6e$)	[PUB]
e_o	67.2 mM/h	max OxPhos ATP rate ($0.4e$)	[LIT]
BASALFUEL	0.40	glucose-independent fuel floor	[FIT]
E_g, E_o	$0.25, 0.12$ mM	glycolysis/OxPhos ADP half-sat	[PUB]
a	168 mM/h	max ATPase rate	[FIT]
A	2.0 mM	ATPase half-saturation for ATP	[PUB]
K_{ak}	0.8	adenylate-kinase equilibrium	[PUB]
k_{ak}	5×10^3/h	adenylate-kinase relaxation rate	[ASM]
		One-carbon / de novo purine	
F_{tot}	0.02 mM	cytosolic folate pool	[ASM]
H	0.01 mM	FTHFS half-sat for formate	[PUB]
K_c	0.002 mM	GART/ATIC half-sat for CHO-THF	[ASM]

Table S2 (continued)

symbol	value	meaning	tag
K_g, K_t	0.01 mM	GART/ATIC half-sat for GAR/AICAR	[ASM]
v_{pur}	0.30 mM/h	resting purine-turnover scale	[ASM]
h	1.2 mM/h	FTHFS capacity ($4v_{pur}$)	[ASM]
k_F	0.625/h	formate exchange coefficient	[PUB]
FMITO	0.5 mM/h	mitochondrial formate capacity	[ASM]
P_{max}, S_{prpp}	0.6 mM/h	PPAT capacity / PRPP supply ($2v_{pur}$)	[ASM]
K_{prpp}	0.3 mM	PPAT half-saturation for PRPP	[ASM]
k_o^{prpp}	2.0/h	lumped PRPP drain	[ASM]
g_{max}, t_{max}	1.5 mM/h	GART/ATIC capacity ($5v_{pur}$)	[ASM]
$k_{\mathrm{deg}}^{\mathrm{int}}$	0.1/h	GAR/AICAR intermediate drain	[ASM]
K_{dP}	0.104 mM	glue PRPP IC_{50} at zero AMP	[FIT]
K_a, n	fitted	glue AMP half-sat / Hill	[FIT]
		Purine branch, degradation, urate	
$k_{amp}^{\mathrm{syn}}, k_{gmp}^{\mathrm{syn}}$	2.0	ADSS / IMPDH mass-action (per *adr*)	[ASM]
V_{max}^{samp}	0.6 mM/h	ADSL capacity (S-AMP→AMP)	[FIT]
K_{samp}	0.02 mM	ADSL half-saturation	[ASM]
k_{imp}^{deg}	1.0/h	IMP degradation	[ASM]
k_{amp}^{deg}	0.5/h	AMP degradation	[ASM]
k_{ade}^{turn}	0.03/h	adenylate catabolism	[ASM]
k_{gnp}^{deg}	0.5/h	GXP degradation	[ASM]
k_{gnp}^{use}	0.3/h	guanine-nucleotide usage	[ASM]
K_{hx}	0.02 mM	xanthine-oxidase half-saturation	[ASM]
k_{xo}	0.6 mM/h	XOR capacity ($2v_{pur}$)	[LIT]
k_{urate}	3.0/h	urate secretion/export	[ASM]
		mTOR / insulin synthesis / secretion	
W_{ATP}	0.03	ATP weight in mTOR purine signal	[ASM]
K_{mp}, h_{mp}	0.12 mM, 2.0	mTOR purine-sensing half-sat / Hill	[ASM]
K_{ec}, h_{ec}	$0.85, 4.0$	energy-charge gate half-sat / Hill	[ASM]
aa, K_{aa}	$2.0, 0.5$ mM	amino-acid pool / translation half-sat	[ASM]
K_G, h_G	7.0 mM, 2.0	glucose insulin-gene half-max / Hill	[FIT]/[LIT]
B_{TX}	0.10	basal insulin-gene drive	[FIT]
B_M	0.29	mTOR-independent insulin fraction	[FIT]
k_{ins}	20 a.u./h	max pro-insulin synthesis rate	[ASM]
k_{mat}	3.0/h	pro-insulin maturation	[LIT]
cost_{ins}	0.02	ATP per a.u. insulin	[ASM]
cost_{pur}	4.0	ATP per mM purine	[LIT]
K_{tr}, h_{tr}	$12.0, 4.0$	secretion-trigger half-sat / Hill (ATP/ADP)	[ASM]
B_{TR}	0.05	basal secretion	[ASM]
k_{sec}	6.0/h	max fractional granule secretion	[LIT]
μ	1/240 /h	pool-renewal dilution	[ASM]

S2.8 Calibration

`recalibrate_all()` fits the added layers in dependency order (energy → IMP branch → mTOR/content → GSIS), each by `least_squares` on log-ratio residuals against independent targets, all evaluated at blood formate 0.3 mM:

1. **Energy** — fit BASALFUEL and ATPase capacity a so ATP/ADP rises 2.4 → 11.6 as

glucose goes $1 \to 10$ mM (Detimary 1998).

2. **IMP branch** — fit the ADSL capacity V_{max}^{samp} so S-AMP rises $\times 3.4$ over glucose $2.8 \to 16.7$ mM (Gooding 2015); the accompanying IMP fall ($\sim -75\%$ vs. target -77%) is structural, not fitted.
3. **mTOR / content** — fit the floor B_M so a full mTORC1 knockout (mtor_mult=0) gives 50% of wild-type insulin content (Ni 2017).
4. **GSIS** — fit the glucose half-max K_G and basal drive B_{TX} to a secretion EC_{50} of 8.0 mM and a stimulation index of 5.5; K_G bounded to $[4, 12]$ mM (glucokinase-linked plausibility).

Absolute per-cell units are set by one scale anchoring the granule pool at stimulatory glucose to the measured β-cell insulin content (20 pg/cell); all calibrated ratios (EC_{50}, stimulation index, formate sensitivity) are invariant under it. Steady states use the pre-integration ($t_{\text{pre}} = 150$ h) plus log-space polish of §S1; trajectory integration uses BDF with rtol $= 10^{-8}$, atol $= 10^{-10}$.

S3 Whole-body loop (Bergman minimal model)

The β-cell is embedded as the insulin source of a Bergman minimal model. The coupled state is $[\, y_{bc}, G_b, I_p, X\,]$: blood glucose G_b, plasma insulin I_p, and remote (interstitial) insulin action X. With effective glucose effectiveness $S_g^{\text{eff}} = S_g(1 + \text{peri})$,

$$\dot{G}_b = R_a(t) - \big(S_g^{\text{eff}} + X\big)G_b + S_g^{\text{eff}} G_{b0}, \tag{S30}$$

$$\dot{I}_p = \beta\, v_{sec}(y_{bc}, G_b, \text{form}_x, f_N) - k_I\, I_p, \tag{S31}$$

$$\dot{X} = -p_2 X + p_2 S_i\,(I_p - I_{b,\text{ref}}), \tag{S32}$$

$$\dot{y}_{bc} = \beta\text{-cell derivatives at glucose } G_b. \tag{S33}$$

Insulin secretion is evaluated at the current blood glucose, closing the glucose→insulin→glucose loop. *Central* formate acts through the β-cell arguments form_x and f_{mito} (raising v_{sec}); the *peripheral* (Carpene) effect enters only as the insulin-independent fractional boost peri to glucose effectiveness. Parameters (per hour): $G_{b0} = 5.0$ mM [LIT], $S_g = 1.5$ [LIT], $p_2 = 1.2$ [LIT], $S_i = 1.0$ [ASM], $k_I = 30$ [LIT], $\beta = 10$ [ASM]; the remote-action gain uses $p_2 S_i$ in place of the classical p_3. Basal insulin is $I_b = \beta v_{sec}/k_I$.

Central-vs-peripheral discriminator. A four-arm intravenous glucose tolerance test (bolus 8 mM, $R_a = 0$, integrated 4 h, BDF $10^{-7}/10^{-9}$) compares a formate-deficient baseline ($f_{\text{mito}} = 0.2\,\text{FMITO}$), a central arm ($f_{\text{mito}} = \text{FMITO}$), a peripheral arm (peri $= 0.30$, tuned to match the central glucose improvement), and both. Outputs are glucose AUC above G_{b0} and total insulin AUC; both arms lower glucose AUC, but only the central arm raises insulin AUC ($> 5\%$ classifies "central-like").

S4 Blood-formate dynamics (diabetes paradox)

To ask how circulating formate itself moves, a dynamic blood-formate pool F_b is added to the fasting whole-body loop ($R_a = 0$), and the β-cell now draws its formate from F_b (closing the formate→insulin→formate loop). A secretion defect $s \in [0, 1]$ scales insulin appearance:

$$\dot{G}_b = -\big(S_g + X\big)G_b + S_g G_{b0}, \qquad \dot{I}_p = \beta\, s\, v_{sec} - k_I I_p, \tag{S34}$$

$$\dot{X} = -p_2 X + p_2 S_i (I_p - I_{b,\text{ref}}), \qquad \text{ins\_frac} = \frac{I_p}{K_I + I_p}, \tag{S35}$$

$$U_{\text{form}} = \big(U_0 + U_{\text{ins}}\, \text{ins\_frac}\big)\frac{F_b}{K_F + F_b}, \qquad \dot{F}_b = D_{\text{diet}} + P_{\text{endog}} - U_{\text{form}} - k_{\text{ren}} F_b. \tag{S36}$$

Blood formate is produced by diet ($D_{\rm diet}$) and endogenous synthesis ($P_{\rm endog}$) and drained by an insulin-scaled, formate-saturating anabolic sink $U_{\rm form}$ (purine synthesis, the main sink) and renal clearance. Parameters (mM, per hour): $P_{\rm endog} = 0.02$, $k_{\rm ren} = 0.5$, $U_0 = 0.01$, $U_{\rm ins} = 0.12$, $K_F = 0.05$ [all ASM]; the insulin half-sat K_I is pinned to the healthy basal plasma insulin (sink half-engaged at health). Dietary intakes (mM/h): low 0.010, normal 0.030, high 0.080. Sweeping the secretion defect $s : 1.0 \to 0.15$ shows that a secretion defect *raises* blood formate (reduced insulin-driven consumption): the overflow route by which incident diabetes can carry higher circulating formate.

S5 Adipocyte model and three-compartment obesity/diabetes coupling

S5.1 Adipocyte (ten ODEs)

The adipocyte reuses the energy and one-carbon/purine machinery of the β-cell but replaces insulin peptide synthesis with insulin-driven lipid storage, and is XOR-rich (a urate source). The state and its initial condition are

$$[\,atp, adp, amp, form, cho, imp, gnp, hx, urate, lipid\,],$$

$$Y_0 = [\,2.0, 0.2, 0.03, 0.3, 0.01, 0.05, 0.1, 0.02, 0.05, 5.0\,].$$

Insulin-dependent GLUT4 uptake, the formate sink, the XOR/urate arm and lipogenesis are:

$$\text{ins\_glut} = B_{GLUT} + (1 - B_{GLUT})\frac{I}{K_{Iglut}+I}, \qquad \text{uptake} = \text{ins\_glut}\,\frac{G}{K_{glut}+G}, \tag{S37}$$

$$v_{fthfs} = h_A \frac{form}{H+form}\frac{\text{thf}}{F_{tot}}, \qquad v_{xo} = \text{xor\_mult}\; k_{xo,a}\frac{hx}{K_{hx}+hx}, \tag{S38}$$

$$v_{lipo} = k_{lipo}\frac{I}{K_{Ilipo}+I}\,\text{uptake}, \tag{S39}$$

$$v_{denovo} = V_{purA}\frac{prpp_a}{K_{prpp,a}+prpp_a}\,\text{onec}\,(0.3 + 0.7\,\text{uptake}), \tag{S40}$$

with hypoxanthine also exported/salvaged ($v_{hx}^{\rm exp} = k_{hxexp}\,hx$, competing with XOR) and urate secreted to blood ($k_{urate,a}\,urate$). Selected parameters: $e_{adip} = 1.2m = 100.8$, $K_{glut} = 5.0$ mM, $K_{Iglut} = 0.5$, $B_{GLUT} = 0.10$; $v_{pur,a} = 0.30$, $h_A = 1.2$, $V_{purA} = 0.6$, $prpp_a = 0.3$ mM; $k_{xo,a} = 0.9$ mM/h (adipose is XOR-rich, Tsushima 2013), $k_{hxexp} = 6.0$/h, $k_{urate,a} = 3.0$/h; $k_{lipo} = 4.0$, $K_{Ilipo} = 0.5$, $k_{lps} = 0.2$/h, $\text{cost}_{lip} = 0.03$; $\mu_a = 1/240$/h [all ASM except $k_F = 0.625$ PUB]. Two one-carbon units are consumed per purine ($\dot{cho} = v_{fthfs} - 2v_{denovo}$).

S5.2 Three-compartment coupling

The full state is [β-cell (15), adipocyte (10), G_b, I_p, X, F_b, U_b] with blood glucose, plasma insulin, remote action, blood formate and blood urate. The β-cell and adipocyte are integrated as single cells whose net blood exchange is scaled by adipose mass M_{adip} and a tissue$\to$blood coupling $c_{adip} = 0.16$. Blood formate and urate obey

$$S_g^{\rm eff} = S_g \cdot \text{sg\_mult}, \qquad \dot{G}_b = -(S_g^{\rm eff} + X)G_b + S_g^{\rm eff} G_{b0}, \tag{S41}$$

$$\dot{I}_p = \beta\, s\, v_{sec} - k_I I_p, \qquad U_{\rm syst} = \left(U_0 + U_{\rm ins}\frac{I_p}{K_I+I_p}\right)\frac{F_b}{K_F+F_b}, \tag{S42}$$

$$\dot{F}_b = D_{\rm diet} + P_{\rm endog} - U_{\rm syst} - c_{adip}M_{adip}\,J_F - k_{\rm ren}F_b, \tag{S43}$$

$$\dot{U}_b = c_{adip}M_{adip}\,J_U + U_0^{\rm ur} - k_{\rm ren}^u U_b. \tag{S44}$$

where $J_F = k_F(F_b - \text{form}_{ad})$ is the per-cell adipose formate uptake and J_U the per-cell adipose urate secretion; $U_0^{\rm ur} = 0.06$ mM/h (non-adipose urate), $k_{\rm ren}^u = 0.3$/h. Integrated to $t = 600$ h (BDF $10^{-6}/10^{-9}$); the healthy lean fixed point self-consistently sets K_I to basal insulin.

S5.3 Obesity, diabetes, supplementation and metformin

Obesity acts on three handles at once: adipose mass $M_{adip} = 3$, XOR induction xor_mult = 2, and insulin resistance $S_i \to 0.5\,S_i$. **Diabetes** acts only through the secretion defect $s = 0.40$. Obesity thus lowers blood formate (adipose sink + XOR) and raises urate, while a secretion defect raises blood formate (overflow) — the two axes move formate in opposite directions (Table S3). **Supplementation** sweeps dietary intake $D : 0.01 \to 0.16$ mM/h with the β-cell either formate-replete (f_{mito} = None, default) or formate-deficient ($f_{\text{mito}} = 0.2\,$FMITO$= 0.1$); only the deficient β-cell is rescuable. **Metformin** at dose $d \in [0, 1]$ applies a dual action: a peripheral/AMPK rise in glucose effectiveness sg_mult $= 1 + d$, and an SHMT2-type reduction of both systemic and β-cell mitochondrial formate, $P_{\text{endog}} \to 0.02(1 - 0.6d)$ and $f_{\text{mito}} \to 0.5(1 - 0.6d)$ — so it lowers glucose while also lowering formate and insulin.

Table S3: Committed 2×2 clinical fixed points (`data/obesity_diabetes.csv`), normal diet.

scenario	glucose (mM)	insulin (a.u.)	formate (mg/L)	urate (mg/dL)	formate vs healthy
lean, non-diabetic	4.79	0.82	1.46	3.44	0%
obese, non-diabetic	4.87	0.84	1.25	3.57	−14%
lean, diabetic	6.26	0.46	1.76	3.45	+21%
obese, diabetic	5.67	0.40	1.53	3.61	+5%

S5.4 Gut-microbiome urate cycle

The class `GutUrateCycle` extends the three-compartment model with a gut compartment that degrades blood urate and returns formate to the circulation, closing the delayed positive feedback of the urate-cycle hypothesis (Vazquez, 2020; gut purine metabolism after Kasahara *et al*, 2023). Blood urate is taken up by the gut route at rate k_{gut} (diverting it from renal clearance) and one of the two purine-derived formate carbons is returned as formate, Y_{form} per urate. The gut processing/transit delay is a linear chain of n_{gut} compartments $g_1 \dots g_{n_{gut}}$ with rate $k_g = n_{gut}/\tau_{gut}$ (the linear-chain trick for a gamma-distributed lag of mean τ_{gut}):

$$\dot{U}_b \mathrel{+}= -\,k_{gut}\,U_b, \quad \dot{g}_1 = k_{gut}\,U_b - k_g\,g_1, \quad \dot{g}_i = k_g\,g_{i-1} - k_g\,g_i, \quad \dot{F}_b \mathrel{+}= Y_{\text{form}}\,k_g\,g_{n_{gut}}. \tag{S45}$$

At steady state the chain gives $k_g g_{n_{gut}} = k_{gut} U_b$, so the return flux is $Y_{\text{form}} k_{gut} U_b$ and the delay affects only the dynamics. Because only one of the two formate carbons consumed per purine is recycled, the loop gain is below one: blood formate rises and converges (bounded feedback) rather than running away. Parameters (all [ASM]): $k_{gut} = 0.2$/h, $Y_{\text{form}} = 1$, $\tau_{gut} = 12$ h, $n_{gut} = 3$; the co-produced acetate (→ lipogenesis) is noted but not tracked. Generated by `make_gutcycle.py`.

S6 The eye compartment

The eye is a retinal insulin-producing cell (retinal pigment epithelium, RPE) running the same β-cell machinery, driven by *blood* glucose and *blood* formate, with strictly local insulin (the blood-retinal barrier excludes plasma insulin) and a low endogenous mitochondrial formate, set to 15% of the β-cell value (EYE_FMITO) [ASM], so that blood one-carbon is rate-limiting.

S6.1 Blood-dependent eye and the retinal window

The blood-dependent eye is the β-cell steady state at f_{mito} = EYE_FMITO with local insulin $S = v^{\text{syn}}_{\text{ins}}$. Retinal outcome is scored by an optimal-insulin window: a trophic maintenance arm minus a higher-threshold pathology arm (VEGF/gliosis/neovascularization), both Hill functions of local insulin,

$$\mathrm{maint}(S) = \frac{S^h}{K_m^h + S^h}, \quad \mathrm{patho}(S) = \frac{S^h}{K_p^h + S^h}, \quad \mathrm{health}(S) = \mathrm{maint}(S) - \mathrm{patho}(S), \tag{S46}$$

with $K_m = 1.7$, $K_p = 2.3$, $h = 6.0$ [ASM]. Because $K_p > K_m$, net health is an inverted-U: deficiency at low insulin, insulin-excess pathology at high insulin, with a retinal optimum at $S^\star \approx 1.98$. This provides an eye-toxicity mechanism requiring no complex-IV inhibition. (A separate regeneration index $S^2/(K_{regen}^2 + S^2)$, $K_{regen} = 3.0$, is used only in descriptive plots.)

S6.2 Local formate feed-forward and its regulation

The RPE may also make its own formate, co-induced with the insulin it produces (the mTORC1 → ATF4 program that induces SHMT2/MTHFD2, plus a cone-paracrine, MTHFD1L-rich supply). This is encoded as a feed-forward on the cell's own mTOR signal, capped by the mTORC1 → S6K1 → IRS-1 negative feedback documented in RPE:

$$f_{\mathrm{mito}} = \mathrm{EYE\_FMITO} + g\, m\, \phi(m), \qquad \phi(m) = \frac{K_{S6K}^H}{K_{S6K}^H + m^H}, \tag{S47}$$

where $m = \mathrm{mtor}$, $g = \mathrm{INDUCE\_GAIN} = 4.0$ [ASM], $K_{S6K} = 0.09$ [FIT, tuned so the regulated eye settles at the optimum under physiological formate], and $H = 6.0$ [ASM]. Setting $\phi \equiv 1$ (feedback removed) gives the unregulated runaway; $g = 0$ recovers the blood-dependent eye.

Solver. Because the feed-forward makes f_{mito} depend on the cell's own steady state, this system is solved by integrating to the stable attractor — iterating BDF blocks ($t_{\mathrm{end}} = 200$ h, rtol 10^{-8}, atol 10^{-10}; up to twelve blocks) until the maximum relative derivative falls below 10^{-7} — rather than by root-polishing. A least-squares root find is *unsafe* here: it converges to *any* fixed point, including dynamically unstable ones, and can report spurious high-insulin states; the physically realised steady state is the stable attractor onto which the dynamics settle. §S6.3 shows the attractor is unique.

S6.3 Monostability of the regulated eye

A positive feed-forward can in principle be bistable (a switch with hysteresis). We therefore characterized the regulated eye before using it (Fig. S1). The feed-forward is a scalar self-consistency condition on the local formate f,

$$f = \mathrm{EYE\_FMITO} + g\, \mathrm{mtor}(f)\, \phi\big(\mathrm{mtor}(f)\big), \tag{S48}$$

where $\mathrm{mtor}(f)$ is the eye cell's steady-state mTOR at imposed local formate f. A fixed point is stable when $d(\mathrm{RHS})/df < 1$. Two independent tests agree that the response is **monostable**: (i) the self-consistency condition (S48) has exactly one root at every blood formate (Fig. S1A), and this persists across feed-forward gains from 2 to 20 — five-fold the reference — with no bistability appearing (Fig. S1C), because the S6K1 cap suppresses the loop before a second stable state can form; and (ii) a basin test integrating from initial states spanning 0.2× to 10× converges to a single attractor at every condition (Fig. S1B). The regulated eye is therefore a smooth homeostat (Fig. S1D), not a bistable switch: across physiological blood formate it holds local insulin near the retinal optimum, lifting the deficient low-formate eye toward it, and rises into excess only when supra-physiological (methanol-range) external formate overrides the cap. Any switch-like (circadian) behaviour of RPE insulin must therefore originate in the phagocytosis signal-gate, not in this metabolic loop.

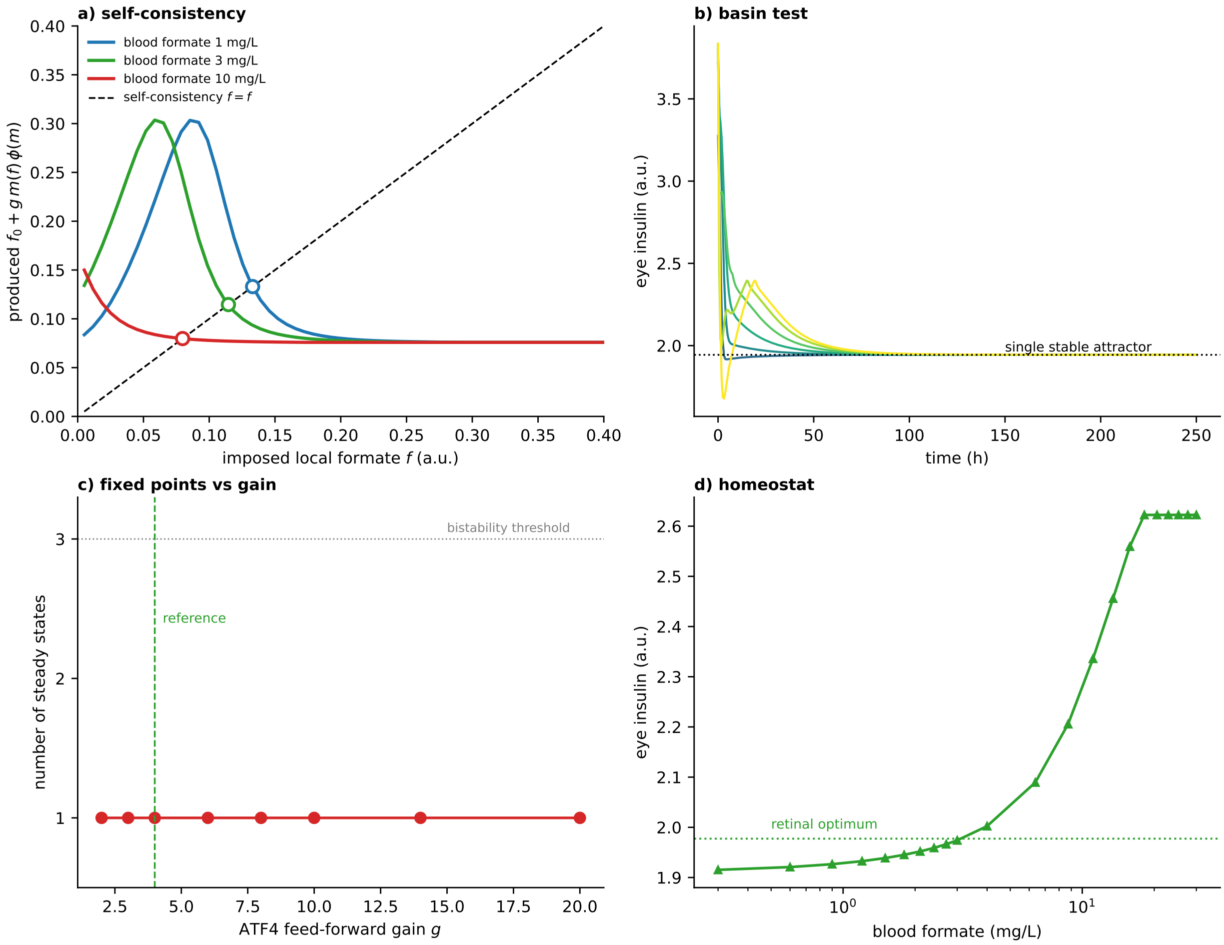

Figure S1: The regulated eye feed-forward is monostable. (A) Self-consistency (S48): produced vs. imposed local formate at three blood-formate levels; each curve crosses the diagonal exactly once (open circles) — a single fixed point. (B) Basin test: trajectories from initial states 0.2× to 10× all converge to one stable attractor. (C) Number of steady states vs. feed-forward gain: monostable throughout, including five-fold the reference gain (dashed) — no bistability. (D) The resulting single-valued homeostat: eye insulin vs. blood formate, buffered near the retinal optimum across the physiological range and rising to excess only above it.

Intravitreal MTHFD2 inhibitor. MTHFD2 is the mitochondrial entry enzyme of the pathway that produces the retina's local formate. An intravitreal MTHFD2 inhibitor is modelled as a factor $\eta \in [0, 1]$ (fraction of activity remaining; $\eta = 1$ uninhibited, $\eta = 0$ full block) scaling the *local* mitochondrial formate only,

$$f_{\text{mito}} = \eta \left[\text{EYE\_FMITO} + g\, m\, \phi(m)\right], \tag{S49}$$

while blood-delivered formate (the barrier-crossing term $k_F(\text{form}_x - form)$ inside the β-cell derivatives) is untouched. Sweeping η shows the therapy is conditional: for an insulin-*excess* retina (lean secretion-defect overflow, hyperglycaemic) partial inhibition (∼70–80%) walks local insulin back through the optimum and net health rises from ∼0.30 to ∼0.42, whereas for a formate-*deficient* retina the same inhibition deepens the deficiency (health ∼0.42 → 0.19). Against a methanol formate flood the inhibitor has no effect (the toxic formate is blood-borne, bypassing MTHFD2): eye insulin and health are identical with and without the drug at intoxication levels. This is generated by `make_mthfd2.py`.

Insulin-production (mTOR) inhibitor. Insulin synthesis is mTOR-driven, so an mTORC1 inhibitor (rapamycin/sirolimus, which reduces β-cell proinsulin biosynthesis) is modelled by the β-cell mtor_mult < 1 threaded through `eye_state_ff`; it caps the mTOR-driven insulin translation *downstream* of formate and purine sensing (and relaxes the S6K feedback and local feed-forward, which are also mTOR-scaled). Unlike the MTHFD2 inhibitor, it therefore acts on a blood-borne overload: under a methanol formate flood a $\sim$50% mTOR inhibition holds eye insulin at the optimum and net retinal health at its peak ($\sim$0.42) across the whole methanol range (versus the $\sim$0.24 insulin-excess floor untreated), with a therapeutic window (peak near 50%; over-inhibition overshoots into deficiency) and little effect on the physiological eye (a proportional cap preferentially clips the saturated excess). Generated by `make_mtor_inhibitor.py`.

Methanol. The methanol overload analysis uses this insulin-excess route alone (complex-IV inhibition switched off): raising blood formate to the symptomatic (> 50 mg/L) and lethal (> 500 mg/L) range drives local insulin past the optimum into pathology. The regulated eye is protected across the physiological range (health at its peak ~ 0.42) but, once external formate floods in, both the regulated and blood-dependent eyes collapse to the same insulin-excess health floor (~ 0.24) — an acute overload that overwhelms regulation, not a chronic mis-set of local insulin.

S7 Dietary meta-analysis mapping

Per-serving formate contents are mapped to model dietary intake and each food's blood-formate prediction (`make_diet.py`). Whole-food contents (creatine 1000 mg, red meat 700 mg, fruit juice 70–600 mg/L) are from *The Spice of Life* (p. 53); caffeine/aspartame drinks and medications use the formate-generation potential of Pietzke, Meiser & Vazquez (2020) — the formaldehyde released by caffeine/aspartame demethylation, taken to full conversion to formate (formate per serving = formaldehyde$_{\mathrm{mmol}}\times$46.03), giving e.g. diet cola $\sim$43 mg, brewed coffee $\sim$31 mg, regular cola $\sim$8 mg, caffeine pill $\sim$47 mg. A serving's formate is treated as a daily intake, $D_{\mathrm{diet}} = 3\times10^{-5}\times(\mathrm{mg})$ mM/h (so 1 g/day maps to the normal intake 0.030 mM/h), and run through the lean-healthy fixed point of §S5. These are compared against pooled relative risks of incident type 2 diabetes per serving (coffee, sugar-sweetened and diet cola, 100% fruit juice, red meat, creatine). The correlation between a food's formate content and its diabetes effect is essentially zero ($r = -0.03$): because the β-cell is formate-replete, dietary formate raises blood formate but leaves plasma insulin and fasting glucose flat, and the model's only appreciable dietary response is the modest, sub-optimal rise in eye insulin.

References

1. Vazquez A. An allosteric metabolite glue couples AICAR inversely to one-carbon availability (submitted) — the Formate–NUDT5 purine/energy core reused here.
2. Witus *et al* (2026) Metabolite glues as a means of purine sensing and chemotherapeutic response. *Nature* (the PPAT–NUDT5 PRPP/AMP glue data).
3. Detimary P *et al* (1998) The changes in adenine nucleotides measured in glucose-stimulated rodent islets. *J Biol Chem* 273:33905.
4. Gooding JR *et al* (2015) Adenylosuccinate is an insulin secretagogue derived from glucose-induced purine metabolism. *Cell Rep* 13:157.
5. Ni Q *et al* (2017) Raptor regulates functional maturation of murine beta cells. *Nat Commun* 8:15755.
6. Tsushima Y *et al* (2013) Uric acid secretion from adipose tissue and its increase in obesity. *J Biol Chem* 288:27138.
7. Kasahara K *et al* (2023) Gut bacterial metabolism contributes to host global purine homeostasis. *Cell Host Microbe* 31:1066–1081.
8. Vazquez A (2020) The urate cycle and obesity. Preprint (ResearchGate 341427356).
9. Pietzke M, Meiser J, Vazquez A (2020) Formate metabolism in health and disease. *Mol Metab* 33:23–37.